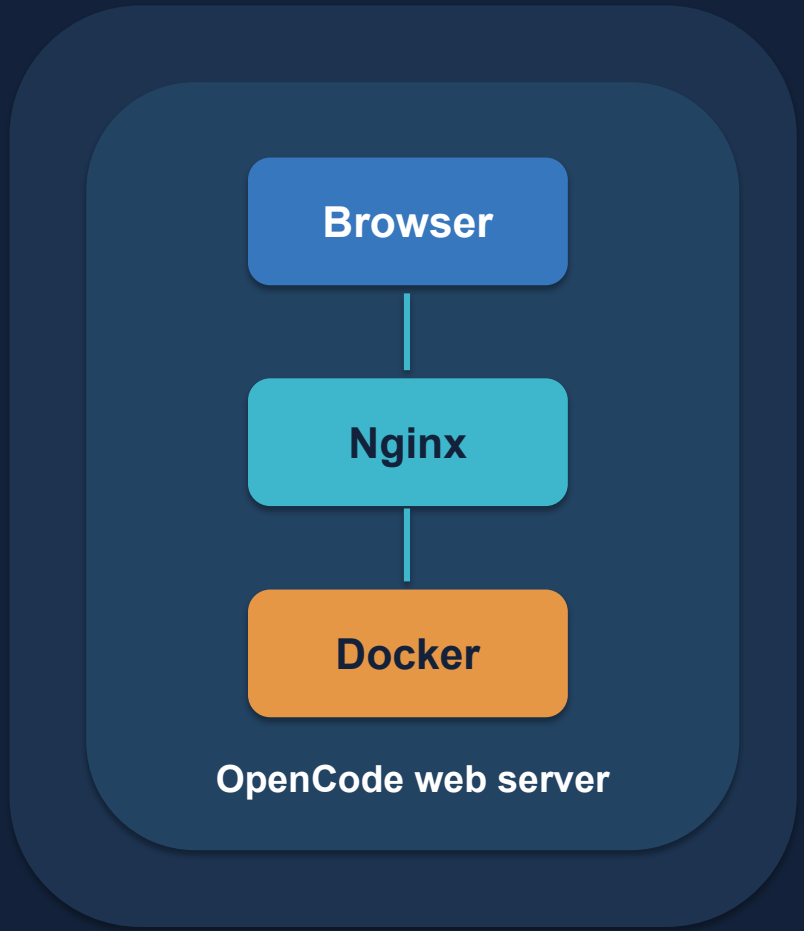
DEPLOYMENT BASICS

From code to a web application

Docker • Dockerfiles • Docker Compose • Nginx



Why do we need deployment?

The goal: anyone on the CMU network can open a browser and use your team's OpenCode instance.

01

Your code

A modified OpenCode application lives in a Git repository.

02

A server runs it

The VM runs the web server even when you disconnect from SSH.

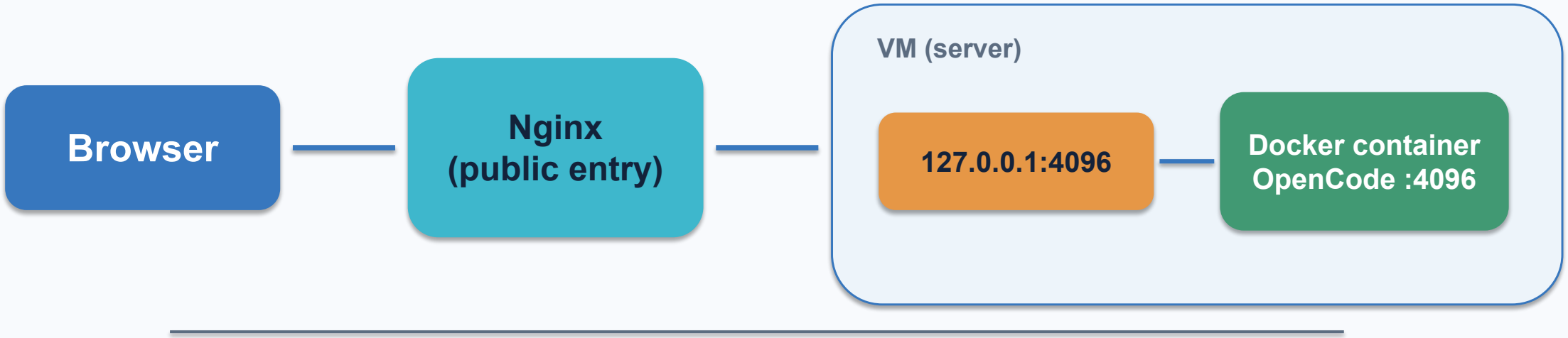
03

People access it

A browser request reaches your app through a public entry point.

Deployment connects the application you built to a server and a route that users can reach.

The big picture: follow one request (P3)



Response travels back through Nginx to the browser

Nginx forwards the request to the VM's local port; Docker maps that port to port 4096 inside the container.

Key idea: Nginx is public-facing; OpenCode stays bound to the VM's local interface.

The big picture: follow one request (For now)

The writeup's request flow, from the user's browser to OpenCode and back.



Docker: package the app's environment

A container is where the OpenCode web server runs on the VM.

Docker image

A packaged template containing the app, dependencies, and runtime environment.

Container

A running instance of that image. It runs OpenCode and listens on port 4096.

Mental model **Image = packaged recipe/result** • **Container = running application**

Dockerfile: describe how to build the image

You write a Dockerfile in your repository; Docker uses it to build your team's image.

```
FROM oven/bun:1.3.14-slim
WORKDIR /workspaces/opencode
COPY . .
RUN bun install
CMD ["bun", "run", "packages/opencode/src/index.ts", "web",
"--hostname", "0.0.0.0", "--port", "4096"]
```

FROM	Choose a base environment.
WORKDIR	Set the working directory inside the image.
COPY / RUN	Copy the project and prepare dependencies.
CMD	Define the command that starts OpenCode.

The Dockerfile defines the image; it does not by itself keep the app running in the background.

Docker Compose: configure and run the service

Compose reads docker-compose.yml and starts the application with the settings you specify.

```
services:
opencode-<ANDREW-ID>:
    build: .
    ports:
  - "127.0.0.1:4096:4096"
restart: unless-stopped
```

build: .

Build the image using the Dockerfile in this directory.

127.0.0.1:4096:4096

Map VM-local port 4096 to container port 4096.

restart: unless-stopped

Restart after an unexpected stop or Docker restart—unless you explicitly stopped it.

```
docker compose up --build -d
```

--build rebuilds the image first • -d runs it in the background

Nginx (For P3): the public reverse proxy

OpenCode is local-only; Nginx receives browser requests and forwards them to OpenCode.



Why this setup? **Users reach Nginx; OpenCode itself is not directly exposed on the VM's public interface.**

Put the pieces together

The recitation follows this sequence to deploy OpenCode on the provided VM.

1. **Connect** — SSH to the VM and clone your team's repository.
2. **Build** — Write a Dockerfile to define the app environment.
3. **Run** — Use Compose to build and run the container in the background.
4. **Recover** — Enable Docker at boot; use a restart policy for the container.

Target outcome: a browser-accessible OpenCode instance that continues running after you disconnect.